Code No: 113BS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write about error detecting codes and error correcting codes used in digital systems.
b) Convert (i) $68BE_{16}$ to Binary and then Octal (ii) Convert 345_{10} to binary and hexadecimal. [8+7]

2.a) Describe the basic theorems and properties of Boolean algebra.
b) Express the following function in sum of minterms and product of maxterms.
$F(A,B,C,D) = B' D + A' D + BD$ [9+6]

3. Implement the following Boolean function F, together with the don't care conditions *d*, using no more than two NOR gates
$F(A,B,C,D) = \sum (0, 1,2,9,11)$
$d(A,B,C,D) = \sum (8,10,14,15)$
Assume that both the normal and complement inputs are available. [15]

4. Simplify the following Boolean functions, using a four variable Karnaugh map method and implement the simplified function using NOR gates
$F(A,B,C,D) = \sum (2,4,5,6,7,10,13,15)$ [15]

5.a) Discuss the analysis and design procedure and draw the logic diagram of a 4 bit BCD subtractor.
b) Implement the following Boolean function with a multiplexer
$F(A,B,C,D) = \sum (0, 1,3,4,8,9,15)$ [9+6]

6. Design a combinational circuit that converts a 4 bit gray code to 4 bit binary converter using XOR gates and explain its operation with examples. [15]

7.a) When did race condition occurs in asynchronous sequential circuits? Explain the noncritical and critical race conditions.
b) Draw the logic diagram of a 4bit binary ripple down counter using flip-flops that trigger on the negative edge of the clock and explain its operation with truth table and timing diagram. [6+9]

8.a) Explain the memory hierarchy in terms of nature, capacity, access times.
b) Explain in detail about sequential memories with examples. [6+9]

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